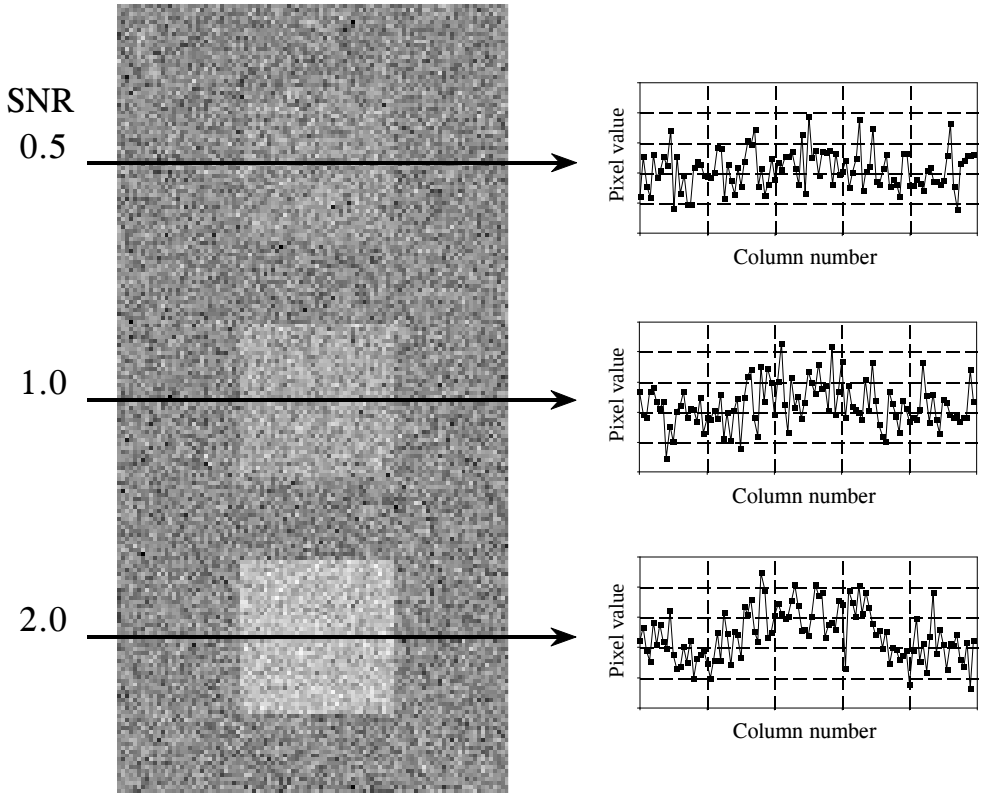

FIGURE 25-8
Minimum detectable SNR. An object is visible in an image only if its contrast is large enough to overcome the random image noise. In this example, the three squares have SNRs of 2.0, 1.0 and 0.5 (where the SNR is defined as the contrast of the object divided by the standard deviation of the noise).

as the brightness of the ambient lightning, the distance between the two regions being compared, and how the grayscale image is formed (video monitor, photograph, halftone, etc.).

The grayscale transform of Chapter 23 can be used to boost the contrast of a selected range of pixel values, providing a valuable tool in overcoming the limitations of the human eye. The contrast at one brightness level is increased, at the cost of reducing the contrast at another brightness level. However, this only works when the contrast of the object is not lost in random image noise. This is a more serious situation; the *signal* does not contain enough information to reveal the object, regardless of the performance of the eye.

Figure 25-8 shows an image with three squares having contrasts of 5%, 10%, and 20%. The background contains normally distributed random noise with a standard deviation of about 10% contrast. The SNR is defined as the contrast divided by the standard deviation of the noise, resulting in the three squares having SNRs of 0.5, 1.0 and 2.0. In general, trouble begins when the SNR falls below about 1.0.